BELLOMASSI

API AND INTEGRATION

Integration Specification (External Systems)

Bellomassi Order-to-Delivery Management Platform

Document ID BM-API-019

Document control

Field	Value
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> **Status labels used in this document:** VERIFIED (confirmed in the current source code or existing documents) · PROTOTYPE (works in the React/LocalStorage prototype with mock or simulated behaviour) · PROPOSED (recommended production design, not built) · PENDING CONFIRMATION (needs a client decision) · NOT IMPLEMENTED · NOT VERIFIED.

Purpose

Specify each external integration — purpose, direction, data, protocol, error handling — and its current status.

Scope

Maps and routing, SMS / WhatsApp, email, OCR, push notifications, SAP, AI providers.

Definitions

Other terms: see **BM-GLO-001** (Project Glossary).

References

- docs/INTEGRATION_GATEWAY.md
- Driver App GPS Implementation Guide §6, §10, §12
- **BM-API-001** (API Specification (Overview and Conventions))
- **BM-ARC-001** (High-Level Design (HLD))

- **BM-SEC-009** (File Upload, OCR, OTP, Email and OpenWA Security)
- **BM-RMP-004** (Integrations Roadmap — SAP, Email, WhatsApp (OpenWA), OCR, Maps and SMS)

1. Integration overview

Integration	Purpose	Direction	Prototype	Production proposal	Status
Google Maps Platform (JS, Geocoding)	Maps, customer geocoding	Outbound	Static demo map	Maps JS in web / RN apps; geocode customer locations	PROPOSED
Google Routes API	Route order, ETA with traffic, VIP time warnings	Outbound	Sequence by priority + pincode	Optimise waypoints per dispatch; ETA per stop	PROPOSED
SMS provider	OTP, delivery note link	Outbound + delivery webhook	OTP shown in app	Transactional SMS with sender ID	PROPOSED / PENDING CONFIRMATION
WhatsApp	OTP, delivery note, notifications	Outbound + webhook	Mock OpenWA session	WhatsApp Business Platform (official) recommended; OpenWA only if client accepts the risk	PENDING CONFIRMATION
Email	Notifications, invoices, documents	Outbound (+ inbound IMAP optional)	Demo mode (SENT_ DEMO)	Transactional provider or SMTP relay with SPF / DKIM / DMARC	PROPOSED
OCR	Extract fields from licences, RC, insurance	Outbound	MOCK provider	Server OCR (Tesseract / cloud OCR) via worker	PROPOSED
Push (FCM / APNs)	Driver and customer push	Outbound	None	Firebase Cloud Messaging / APNs	PROPOSED
SAP	Items, stock, customers, credit, sales orders, deliveries	Bidirectional	None	Stage 2 — interface to be confirmed (OData, RFC/BAPI or IDoc)	PROPOSED
AI providers	Assistant answers	Outbound	DEMO engine	Optional; only with data-protection approval	NOT VERIFIED

2. Common integration design — PROPOSED

- **Outbox pattern**: business transaction writes an outbox row; a Celery worker sends it; at-least-once with idempotent providers keys.
- **Retries**: exponential back-off (1, 5, 15, 60 min), then dead-letter with alert.
- **Logging**: request / response metadata (no secrets, masked phone numbers) in `messages` / integration log.
- **Secrets**: provider keys in a secrets manager; never in the browser.
- **Sandbox**: every adapter has a mock implementation (as in the prototype) used in tests and demo environments.

3. Message content (customer)

Event	Channel	Template (example)
Driver arrives	SMS / WhatsApp	“Bellomassi: your delivery OTP for order {order_number} is {otp}. Valid {minutes} min. Share it only with the driver.”
Delivery confirmed	SMS / WhatsApp / email	“Delivered: {order_number}. Delivery note {note_number}: {short_link}”
Delivery not completed	SMS / WhatsApp	“We could not deliver {order_number} ({reason}). Our team will contact you to reschedule.”

4. SAP integration (stage 2) — PROPOSED, scope to confirm

Object	Direction	Frequency	Key mapping
Material (item) master	SAP → Bellomassi	Daily + on change	`products.sap_material_no`
Customer master and credit	SAP → Bellomassi	Daily + on change	`customers.sap_customer_no`
Stock per plant / storage location	SAP → Bellomassi	Hourly	`hubs` ↔ plant mapping
Approved sales order	Bellomassi → SAP	Real time	`orders.order_number` ↔ SAP order
Delivery / goods issue	Bellomassi → SAP	On dispatch confirmation	`dispatches` ↔ outbound delivery
Proof of delivery	Bellomassi → SAP	On completion	Delivery note number

5. Existing gateway contract (prototype) — NOT VERIFIED

The prototype defines HTTP endpoints for a gateway (`/email/send`, `/email/imap/messages`, `/openwa/sessions/:name/…`, `/ocr/process`, `/ai/chat`, `/ml/forecast`). They are documented in `docs/INTEGRATION_GATEWAY.md` and can be reused as the internal interface of the integration workers; they have never run against a live gateway.

Revision history

Version	Date	Author	Description
1.0	2026-11-12	Documentation team	Initial draft prepared from analysis of the bellomassi-premium-ui source code and existing project documents.